

MISSION 100

JEE 2027

Lecture - 01

Inorganic Chemistry

Classification of Elements and Periodicity in Properties

Periodic Table

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Topics to be covered



Complete Periodic Table

Introduction

Isotope : Atomic number same but different mass number.

Ex ${}_1\text{H}^1, {}_1\text{H}^2, {}_1\text{H}^3$ ✓

Isobar : Same mass number but different atomic number.

Ex ${}_{18}\text{Ar}^{40}$ and ${}_{20}\text{Ca}^{40}$ ✓

Iso-electronic species : Species (atom, molecules or ions) having same number of electrons are called iso-electronic.

Ex $\text{C}^{4-}, \text{N}^{3-}, \text{O}^{2-}, \text{F}^-, \text{Ne}, \text{Na}^+, \text{Mg}^{2+}, \text{Al}^{3+}$

CO_2 N_3^-
 22 22
 Total e^- s

18 e^- rule,
Complex
 only v-se-

Iso-sters : Species having same number of electrons & same number of atoms.

Ex: $\text{CH}_4, \text{NH}_4^+$

50+ $\rightarrow$ H.W

Iso-diaphers : Species having same difference in number of neutrons and protons

Ex: $_{15}\text{P}^{31}, _{17}\text{Cl}^{35}, _{11}\text{Na}^{23}, _3\text{Li}^7, _{13}\text{Al}^{27}, _{19}\text{K}^{39}, _9\text{F}^{19}$

Main ✓

All have isotopic excess of $(A - 2Z) = 1$.

Isotones : Elements having the same number of neutrons are known as isotones.

Ex: $_{12}\text{Mg}^{24}$ and $_{11}\text{Na}^{23}$

Define the number of electron, proton and neutron in P, D and T

	P	D	T
	${}^1_1\text{H}^1$	${}^2_1\text{H}^2$	${}^3_1\text{H}^3$
$e^- \rightarrow$	1	1	1
$p \rightarrow$	1	1	1
$n \rightarrow$	0	1	2

Question



H.W

70+

Which of the following match the column :
Column-I

Column-II

A



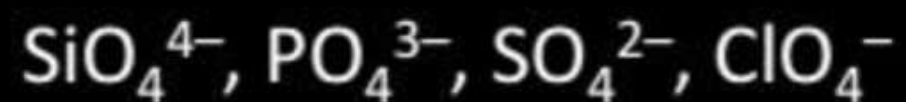
p. Isosters

B



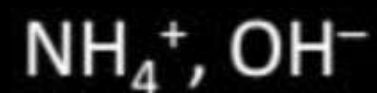
q. Isobar

C



r. Isotope

D



s. Isoelectronic

Quantum Number



Quantum Number

Shell $\rightarrow$ Sub Shell $\rightarrow$ Orbital
 $\downarrow$
 e^-



Define the position the electrons in atom.

Set of 4 mathematical values

Types

1. Principal Quantum number
2. Azimuthal Quantum number
3. Magnetic Quantum number
4. Spin Quantum number



Principal Quantum Number



Denoted by : n

Define : *Size*

Presentation : *K, L, M, N, O, P, ... Shell*

Value : *1 2 3 4 5 6 - - - ?*

$n \uparrow$, Size of shell $\uparrow$

$n \uparrow$, Energy of shell $\uparrow$

$n \uparrow$, Velocity of electron shell $\downarrow$

Question

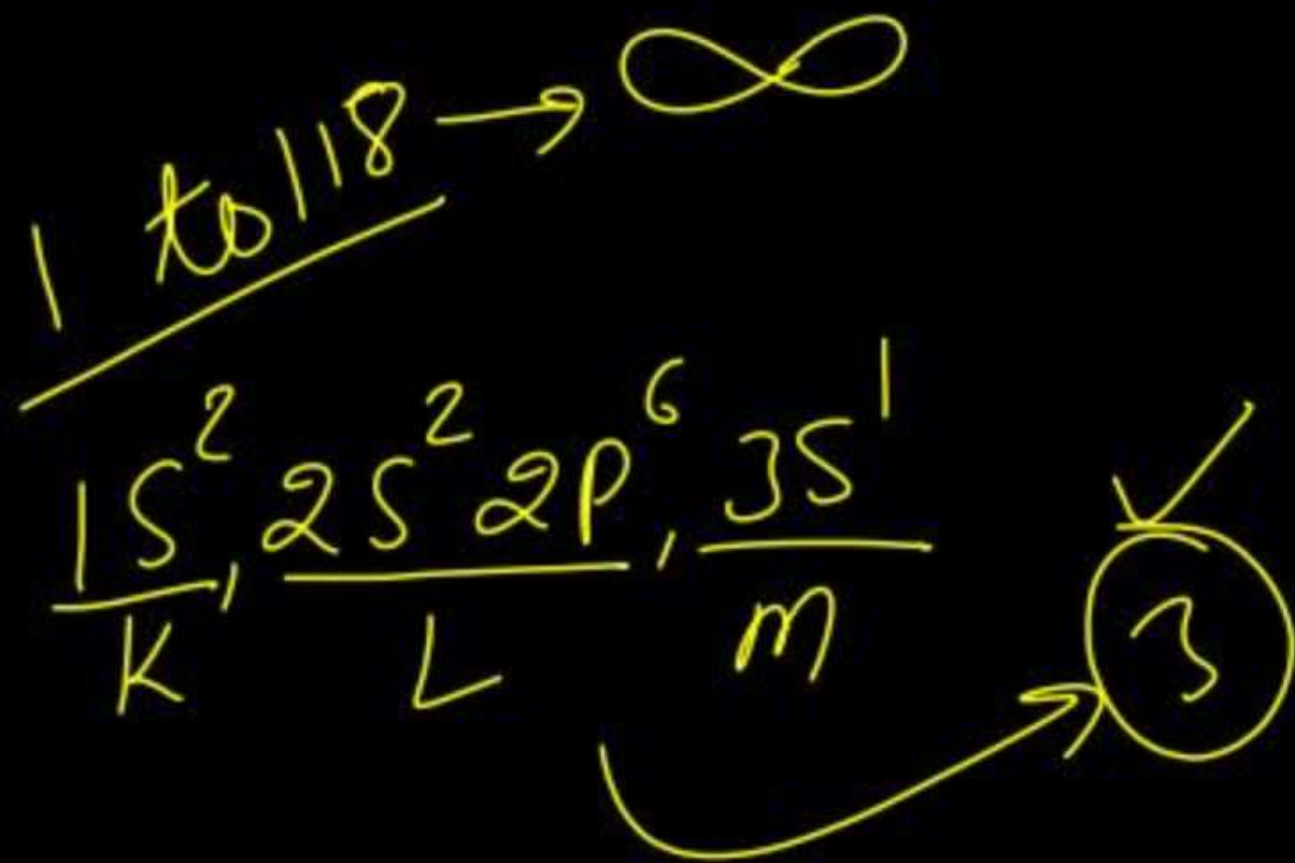
Number of shell in Na

A 3

B 4

C 5

D infinite





Azimuthal Quantum Number



Also known as angular, secondary and subsidiary quantum number.

Denoted by : l

Define : Shape (Sub Shell) s p d f
Complex

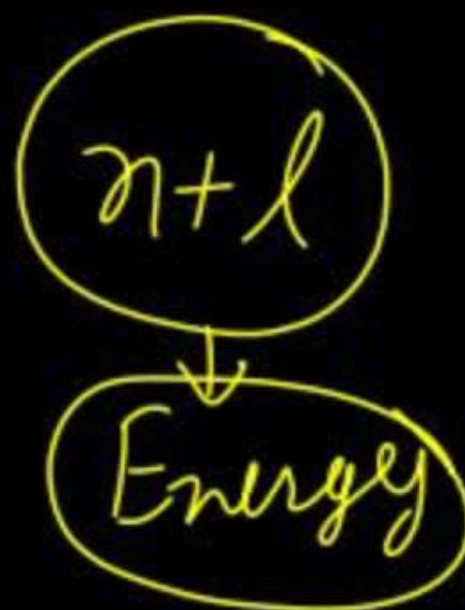
Value : $0, 1, 2, 3, \dots, \infty$
 s, p, d, f

Presentation :

~~Size of sub-shell~~

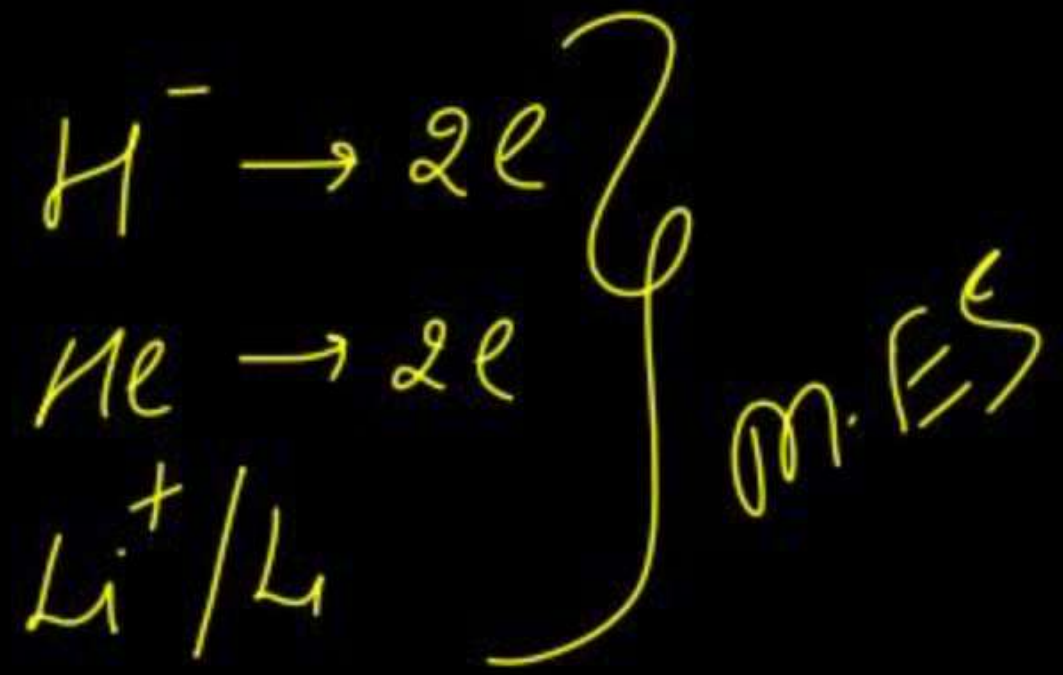
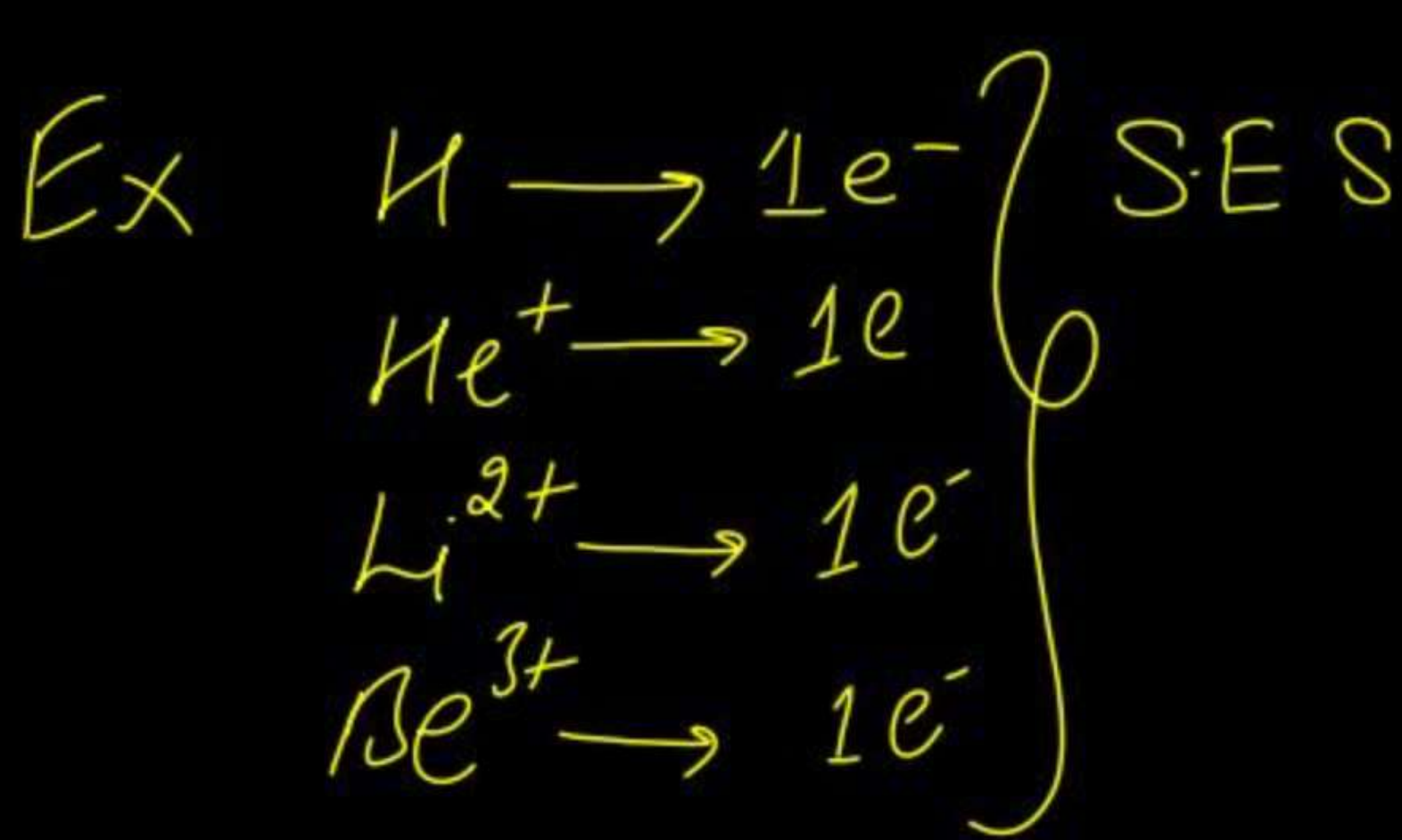
~~Energy of sub-shell~~

Relation in between n and l



$n \rightarrow$ Same

$s < p < d < f$



① S.E.S

② m.E.S (n+1)

only $\textcircled{n} \rightarrow 4$ $4s = 4p = 4d = 4f$

$\textcircled{n} \rightarrow 4$ $4s < 4p < 4d < 4f$

Question

Relation in between n and l

- A** $n > l$ ✓
- B** $n < l$ ✗
- C** $n = l$ ✗
- D** none of these

Handwritten notes:

- $3d$ ✓
- $n \rightarrow 3$
- $l \rightarrow 2$
- $l = 0 \text{ to } n-1$
- $3f$ ✗
- $2d$ ✗
- $1d$



Magnetic Quantum Number

Denoted by : m

Define : *orientation*

Value : $-l$ to $+l$ (0)

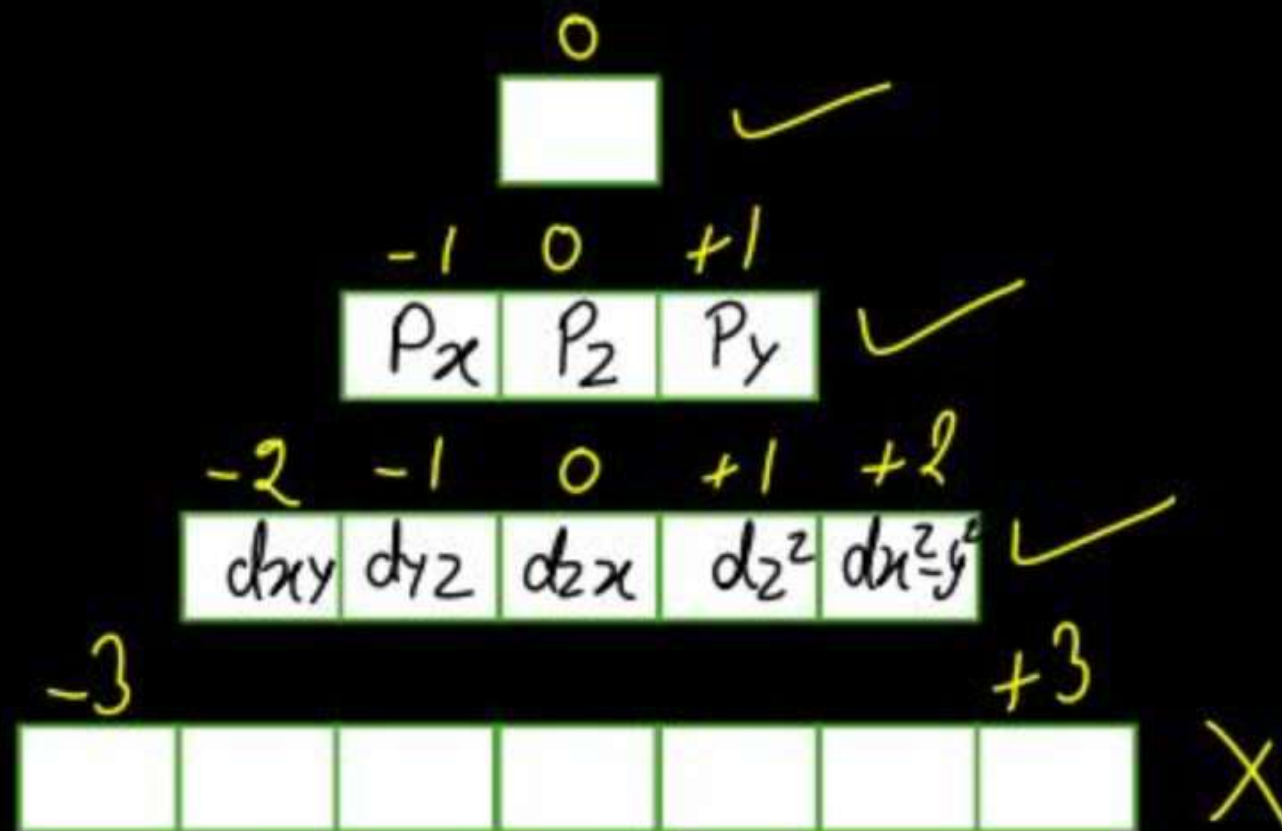
Presentation :

For s $l = 0$

For p $l = 1$

For d $l = 2$

For f $l = 3$



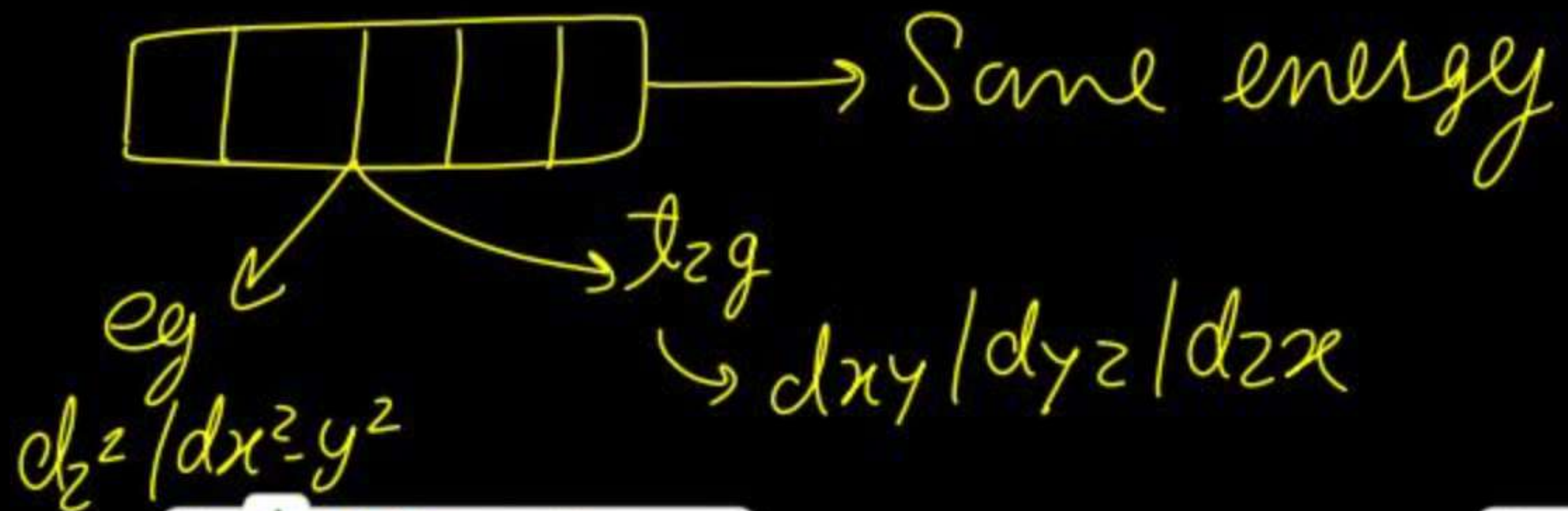
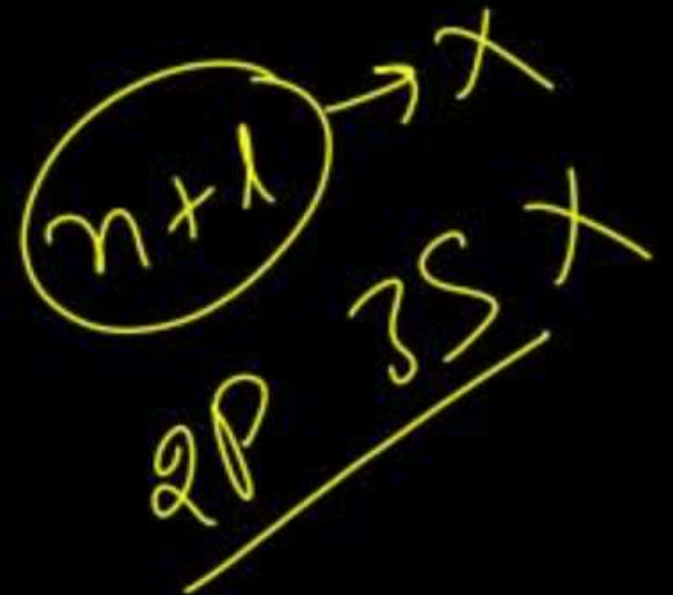
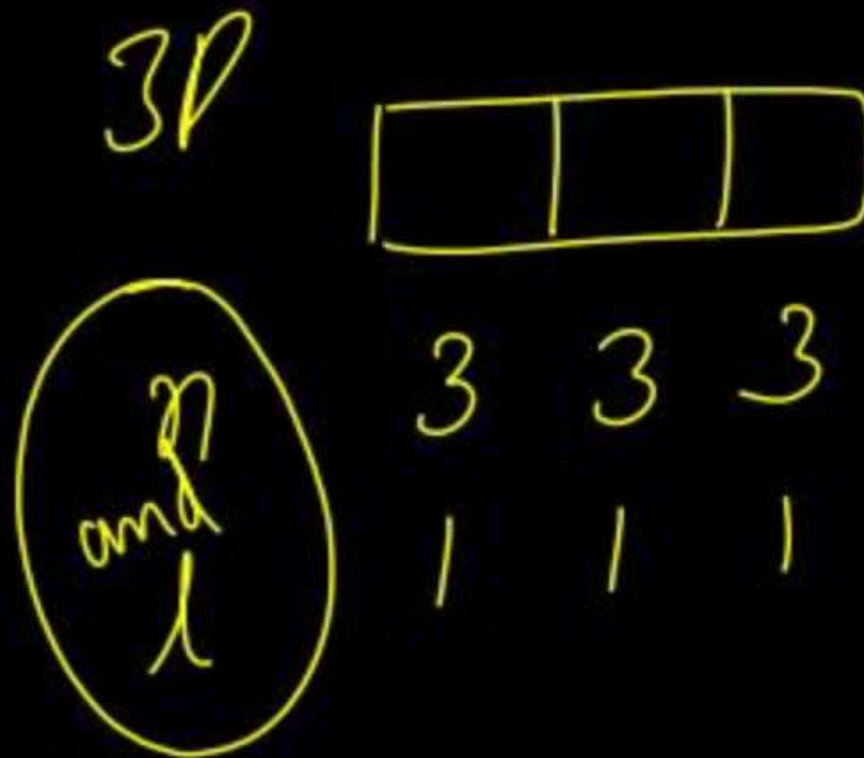


Degenerate orbital

Orbitals have same energy level.

Required condition

1. n and l value must be same
2. Electronic or magnetic field must be absent





Spin Quantum Number

Denoted by : s

Value : $+\frac{1}{2} \quad -\frac{1}{2}$

Define : Spin of electron around the nucleus

Types

(i) Clockwise

(ii) Anticlockwise



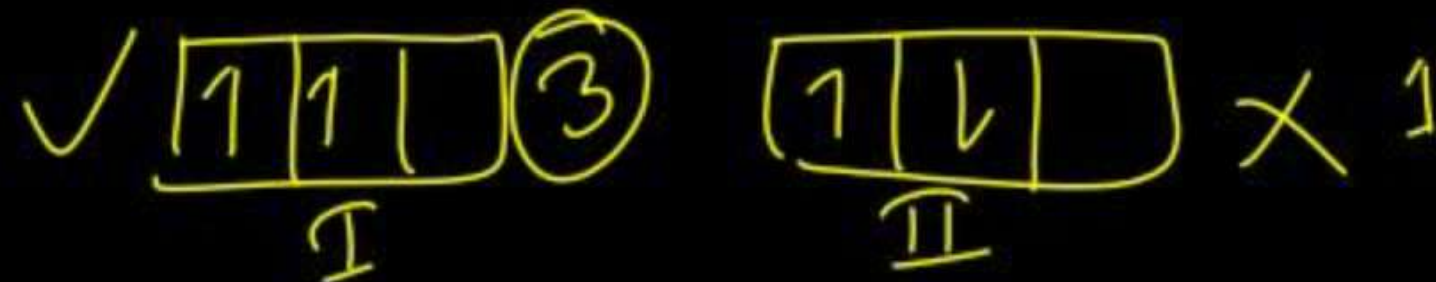
Summary

Shell n.	S. Shell n	Orbitals n^2	e^- $2n^2$
K 1	1(1s)	1	2
L 2	2(2s 2p)	4	8
M 3	3(3s 3p 3d)	9	18
N 4	4(4s 4p 4d 4f)	16	32

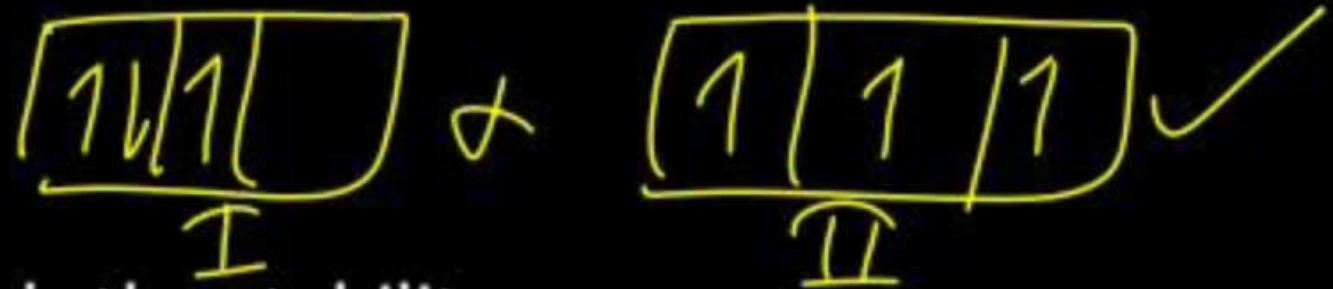


Hund's rule of maximum multiplicity

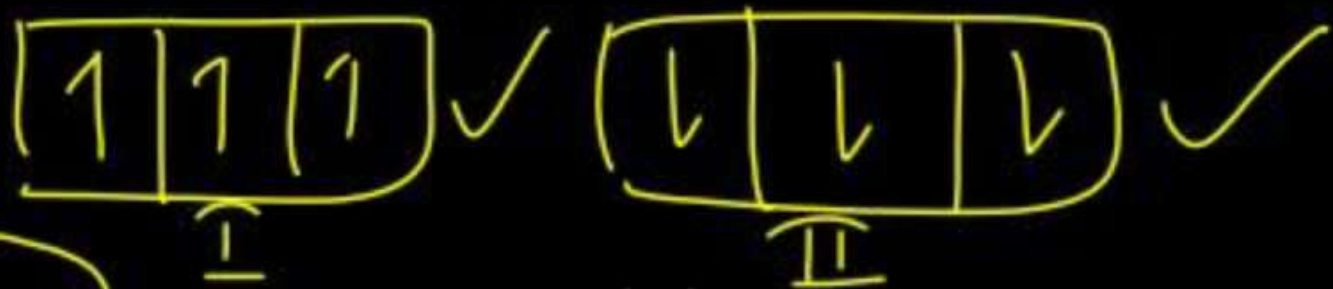
1. Spin of the electrons must be same until they are half field.



2. Pairing of the electrons must be restricted until they are half field.



3. Symmetry leads the stability.



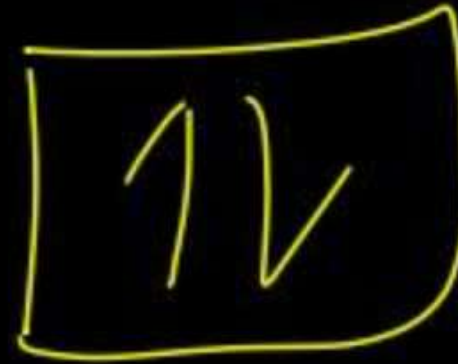
4. More multiplicity, more acceptability

$2|S|+1$
 $S \rightarrow \text{Sum of spin}$



Pauli's exclusion principle

Maximum two electrons of opposite spin can be accommodated in any orbital.



Case I

$(n+1)$

$n+1 \uparrow \quad E \uparrow$

$n+1 \downarrow \quad E \downarrow$

$E \rightarrow \underset{4}{4S} < \underset{5}{4P}$

Case II

$n+1 \rightarrow \text{Same}$

$(n) \rightarrow \text{min } E \downarrow$

$\underset{3}{2P} < \underset{3}{3S}$



Aufbau Principle



Electron filling : From low energy level orbital to high energy level orbital.

n
 n s p
 $n-1$ d
 $n-2$ f

Period

1

1s

2

2s

2p

3

3s

3p

4

4s

3d

4p

5

5s

4d

5p

6

6s

4f

5d

6p

7

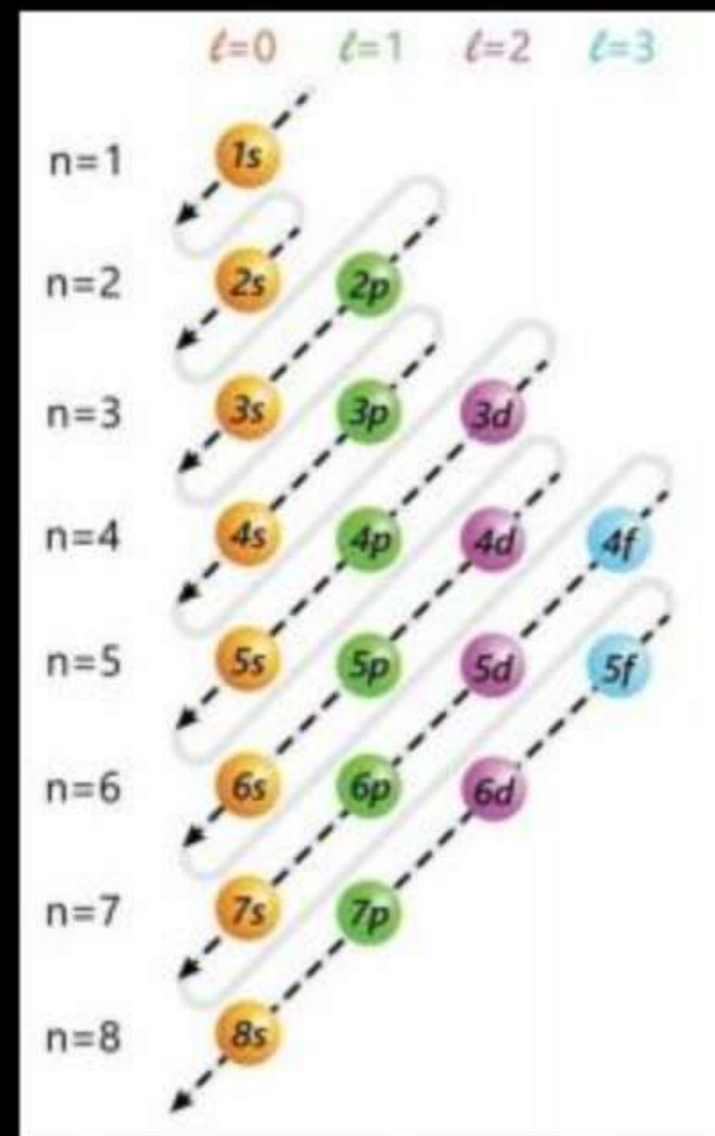
7s

5f

6d

7p

$n+l \uparrow$





Magnetic Behaviour

Two types :

1. Paramagnetic : Unpaired electron is present. ✓



2. Diamagnetic : Unpaired electron is absent.



Magnetic Moment

$$\mu = \sqrt{n(n+2)} \text{ BM}$$

$n \rightarrow$ no of $\uparrow$ v.p.e $\uparrow$ $\uparrow$

Development of Periodic Table

- (A) **LAVOISIER CLASSIFICATION :**
Classification of element. *Metal / Non-Metal*
- (B) **PROUT'S HYPOTHESIS (Unitary theory):**
All the elements are made up of hydrogen.
- (C) **DOBEREINER TRIAD RULE [1817]:**
At. wt. of middle element is equal to the average at. wt. of first and third element.
- (D) **NEWLAND'S OCTAVE RULE [1865]**
The properties of every 8th element was similar to the 1st element. (like musical notation).
- (E) **Cylindrical or Helix Periodic Table :**
A spiral graph called a telluric helix.
- (F) **LOTHER MEYER'S CURVE [1869]**
Graphs in between atomic weight and atomic volume.

main

$$B = \frac{A + C}{2}$$

Which of the following is not a dobereiner triad

- A** Li, Na, K
- B** Mg, Ca, Sr
- C** Cl, Br, I
- D** S, Se, Te

Ans. B

Which of the following set of elements obey Newland's octave rule-

- A** Na, K, Rb
- B** F, Cl, Br
- C** Be, Mg, Ca
- D** B, Al, Ga

Ans. C

Match the following statements with law and their discoverer.

	State		Law		Discover
(a)	Middle element of each of triads had an atomic weight halfway between the atomic weights of the other two.	(i)	Law of triads	(p)	de Chan Courtois
(b)	Arrangement of known elements in order of increasing atomic weights in a cylindrical table of elements to display the periodic recurrence of properties.	(ii)	Telluric screw or Helix law	(q)	Dobereiner
(c)	Arrangement of the lighter elements in increasing order of their atomic weights and noted that every eighth element had properties similar to the first element.	(iii)	Law of octaves	(r)	Newlands

Ans. (a) - i, q ; (b) - ii, p ; (c) - iii, r

In the Lothar Meyer curve, which of the following option is incorrect :

- A** Alkali metals occupied peak position at curve.
- B** Halogens occupied ascending position at curve.
- C** Alkaline earth elements occupy descending position at curve.
- D** Alkali metals are in the lower curve of graph.

Ans. D

(G) MENDELEEV'S PERIODIC TABLE [1869] :



Row	Column Series	I		II		III		IV		V		VI		VII		VIII			0
		A	B	A	B	A	B	A	B	A	B	A	B	A	B				
1	1	H																	He
2	2	Li		Be		B		C		N		O		F					Ne
3	3	Na		Mg		Al		Si		P		S		Cl					Ar
4	4	K		Ca			Sc		Ti		V		Cr		Mn	Fe	Co	Ni	Kr
	5		Cu		Zn	Ga		Ge		As		Se		Br					
5	6	Rb		Sr			Y		Zr		Nb		Mo		Tc	Ru	Rh	Pd	Xe
	7		Ag		Cd	In		Sn		Sb		Te		I					
6	8	Cs		Ba			La*		Hf		Ta		W		Re	Os	Ir	Pt	Rn
	9		Au		Hg	Tl		Pb		Bi		Po		At					
7	10	Fr		Ra			Ac*		Rf										

63

At. Wt.



Merits



(a) Study of elements :

Early

(b) Correction of doubtful atomic weights : Atomic weight = Valence $\times$ Equivalent weight:

U, Be, In, Au, Pt

(c) Prediction of new elements :

Ca, Sc, Ge, Tc

Eka Aluminium :

Eka Boron :

Eka Silicon :

Eka Manganese :

Demerits

(a) Position of hydrogen:

Not fix

(b) Position of isotopes:

Nonfix

C. Anomalous pairs of elements:

Ar	K	Co	Ni	Te	I	Th	Pa
39.9	39.1	58.9	58.6	127.5	127	232	231

THANK YOU

